

DMAIC

The core methodology of Lean Six Sigma — a structured approach to improve existing processes and sustain measurable results

OVERVIEW

DMAIC Within the Business Lifecycle

DMAIC is the continuous improvement engine inside day-to-day operations. While often associated with manufacturing, it is equally valuable in healthcare, finance, cloud computing, logistics, chemical engineering, and AI.

1

Customer Need

Business strategy is shaped by what customers require

2

Business Process & Operations

Strategy is executed through defined business processes

3

Performance Monitoring

Operations are tracked against measurable KPIs

4

DMAIC Improvement

Continuous improvement drives higher customer value

The key difference between DMAIC and a general project lifecycle is that **DMAIC focuses on improving an existing process**, whereas frameworks such as PMP focus on delivering a project from initiation to completion.

REAL-LIFE SCENARIO

AWS Data Center — Cooling Energy Reduction

Amazon Web Services seeks to reduce unnecessary cooling energy consumption in a data center while maintaining full server reliability.



Cooling Energy

Current: **25 MW** → Target: **22 MW**



Annual Cooling Cost

Current: **\$18 million**



Estimated Savings

Target: **\$3 million** annually



Business Objective

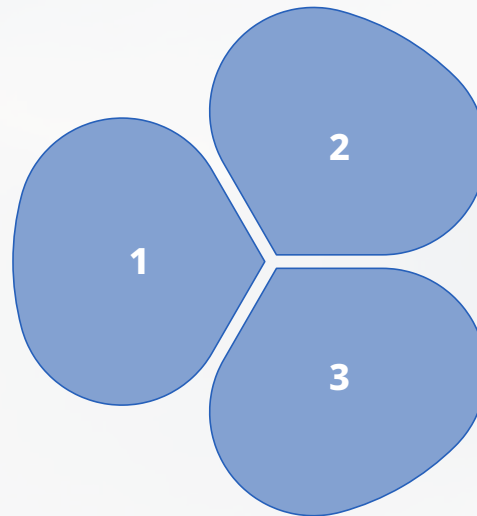
Reduce cooling energy by **12%** without affecting uptime

PHASE 1

Define

Business Problem

Cooling systems consume more energy than necessary because equipment runs at conservative setpoints



Business Impact

Higher electricity costs, increased water usage, reduced sustainability, and lower profit margins

Deliverables

Problem Statement, SIPOC Diagram, Voice of Customer (VOC), Project Charter, Stakeholder Register

What is the problem?

Who is affected?

Why is it important?

What defines success?

Measure

Collect reliable data before making changes



Data Sources

SCADA, BMS, EPMS, Weather, Server utilization, Cooling tower performance



Statistical Tools

Mean, Standard deviation, Histograms, Control charts, Process capability



Business Analytics

Power BI dashboards: daily energy, monthly trends, equipment utilization, PUE, cost

Example Measurements

Metric	Current Value
Cooling Power	25 MW
Server Temperature	22°C
Outdoor Temperature	34°C
Water Usage	3.5M gal/month
PUE	1.38

Analyze



People

Manual overrides and different operating practices across teams



Process

Fixed cooling schedules with no dynamic optimization



Equipment

Dirty coils, sensor drift, and aging fans reducing efficiency



Environment

Seasonal weather variations and humidity affecting load

Key Insight from Regression Analysis

Outdoor temperature explains only **35%** of energy variation

Server utilization explains **70%** of energy variation

Cooling is driven more by IT load than by weather — this insight fundamentally changes the optimization strategy.

Analysis Methods

Fishbone Diagram, Five Whys, Pareto Analysis, Correlation Analysis, Regression, Hypothesis Testing

PHASE 4

Improve

01

Engineering Solutions

Adjust chilled water temperature setpoints, improve airflow management, and optimize fan speeds dynamically

03

Pilot Results

Cooling Power: 25 MW → 22 MW | Water Usage: 3.5M → 3.0M gal |
Annual Cost: \$18M → \$15M | PUE: 1.38 → 1.24

02

AI Optimization

Train an XGBoost model to predict optimal cooling setpoints based on real-time server utilization data

04

Business Value

\$3M annual savings, lower carbon emissions, better sustainability metrics, and improved operational efficiency



PHASE 5

Control

Prevent Regression — Make Gains Permanent

The gains become part of normal operations instead of a one-time improvement.

Automated Dashboards

Real-time monitoring of Cooling Power, PUE, Water Usage, and Equipment Health

AI Alerts & SOPs

Automatic alerts if cooling power exceeds 22 MW; operator training and standard procedures

KPI Reviews

Weekly KPI reviews and preventive maintenance schedules to sustain improvement

CROSS-CUTTING ROLE

Business Analytics Throughout DMAIC

Business analytics transforms operational data into actionable decisions at every phase.

DMAIC Phase	Business Analytics Contribution
Define	Quantify business impact and establish KPIs
Measure	Build dashboards and collect operational data
Analyze	Discover root causes using statistics and machine learning
Improve	Model scenarios and estimate ROI
Control	Monitor KPIs and detect deviations automatically

Engineering Throughout DMAIC

Engineering ensures that proposed improvements are technically feasible and safe.

DMAIC Phase	Engineering Contribution
Define	Understand system requirements and constraints
Measure	Install and validate sensors; verify instrumentation
Analyze	Apply thermodynamics, fluid mechanics, and process engineering
Improve	Design and implement technical solutions
Control	Standardize procedures and monitor system performance

PMP Process	DMAIC Relationship
Initiating	Define
Planning	Measure + Analyze
Executing	Improve
Monitoring & Controlling	Control
Closing	Lessons Learned and project closure

DMAIC identifies *what should change* and *why*; PMP provides the structure to implement those changes across teams, budgets, schedules, and stakeholders.

End-to-End Value Creation

How modern organizations convert problems into business value



Engineering

Provides the technical solution — ensuring improvements are feasible, safe, and effective



Business Analytics

Identifies opportunities and measures outcomes at every stage of the process



AI & Machine Learning

Enhances prediction and optimization beyond traditional engineering methods



PMP Execution

Coordinates implementation so improvements reach teams, budgets, and stakeholders

DMAIC systematically improves processes so that technical achievements become measurable business value — and ultimately, greater customer satisfaction.